

OPPDAL-SETER, Norway

WMO: 012450

Lat: 62.604N

Lon: 9.667E

Elev: 1988

StdP: 13.67

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.2	(c) 5.9	(d) -6.9	(e) 4.1	(f) 8.2	(g) -3.6	(h) 4.9	(i) 10.1	(j) 22.8	(k) 28.8	(l) 20.6	(m) 29.9	(n) 2.5	(o) 280	(p) 0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	74.3	58.8	70.1	56.8	67.1	55.5	60.8	70.5	58.8	67.9	56.9	65.1	6.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.4	73.1	64.7	54.6	68.3	62.4	52.8	63.8	60.6	27.9	70.3	26.5	68.0	25.1	65.0	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	16.1	13.8	-2.7	79.4	4.8	4.1	-6.2	82.3	-9.0	84.7	-11.7	87.0	-15.2	90.0
			-3.5	63.7	4.6	2.1	-6.8	65.2	-9.5	66.5	-12.1	67.7	-15.5	69.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.3	24.8	27.1	29.7	36.4	45.4	50.5	56.3	54.1	49.1	39.6	30.9	27.1		
	DBStd	13.34	8.59	8.19	7.48	7.55	8.36	5.84	6.83	5.20	5.94	6.78	9.36	9.49		
	HDD50	4462	782	640	628	412	197	66	18	19	88	328	575	709		
	HDD65	9386	1247	1060	1093	858	608	436	281	340	477	788	1024	1174		
	CDD50	562	0	0	0	4	54	80	214	144	61	5	0	0		
	CDD65	13	0	0	0	0	0	0	12	1	0	0	0	0		
	CDH74	135	0	0	0	0	8	1	121	5	0	0	0	0		
	CDH80	18	0	0	0	0	0	0	18	0	0	0	0	0		
	WSAvg	6.2	6.4	6.7	7.3	6.5	6.5	6.5	5.7	5.2	5.4	5.8	5.7	6.5		
	PrecAvg	24.2	1.9	1.8	1.9	1.2	1.2	2.4	3.1	3.2	1.9	1.8	1.8	2.0		
(15) Precipitation	PrecMax	34.5	5.9	4.0	5.3	5.3	2.5	5.5	4.8	7.0	5.2	4.8	6.5	4.7		
	PrecMin	18.3	0.0	0.0	0.3	0.1	0.4	0.6	1.5	1.2	0.5	0.4	0.1	0.4		
	PrecStd	3.7	1.5	1.1	1.3	1.0	0.5	1.3	0.9	1.4	1.1	1.0	1.7	1.1		
	PrecStd	3.7	1.5	1.1	1.3	1.0	0.5	1.3	0.9	1.4	1.1	1.0	1.7	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.3	48.3	49.1	61.5	74.1	71.9	83.0	73.9	69.3	63.4	53.6	49.0		
		MCWB	38.1	39.3	40.0	47.7	56.8	55.7	62.6	57.5	55.9	55.1	45.0	42.6		
	2%	DB	41.3	44.1	45.2	56.6	68.3	68.3	77.3	69.3	65.3	54.8	49.4	45.0		
		MCWB	35.7	37.4	38.4	44.4	53.7	54.7	60.8	56.0	53.7	45.7	42.3	39.1		
	5%	DB	38.7	40.3	43.1	52.2	64.5	64.8	72.8	66.3	61.9	52.2	45.8	42.1		
		MCWB	33.8	34.3	36.9	42.4	51.2	53.2	59.2	55.6	52.5	44.5	39.9	36.9		
	10%	DB	36.7	37.5	40.5	48.2	60.5	61.4	68.6	63.5	59.1	49.5	42.9	39.0		
		MCWB	32.1	32.5	34.9	39.5	49.5	51.8	57.9	54.3	50.8	42.7	37.5	34.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	40.4	42.0	48.6	58.0	59.4	64.5	61.2	57.4	54.8	45.7	43.1		
		MCDB	44.7	47.4	46.9	60.2	71.9	68.1	77.8	68.0	65.8	63.0	52.7	48.6		
	2%	WB	36.5	37.3	39.4	45.3	54.8	56.7	62.3	58.5	55.3	47.6	42.9	39.8		
		MCDB	40.4	43.2	44.2	54.8	66.3	65.5	74.3	66.0	63.6	53.7	48.6	44.5		
	5%	WB	34.4	34.9	37.4	42.9	52.3	54.4	60.4	56.7	53.2	45.2	40.5	37.2		
		MCDB	38.2	39.5	42.6	51.5	62.8	62.3	70.4	63.6	60.5	51.6	45.6	41.8		
	10%	WB	32.4	32.9	35.3	40.5	49.9	52.3	58.5	54.9	51.5	42.8	37.7	34.3		
		MCDB	36.2	37.3	40.1	47.9	59.6	59.7	67.7	61.9	58.3	48.7	42.5	38.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	10.3	12.9	13.8	15.2	15.4	16.1	14.8	13.6	12.1	9.8	10.4		
		MCDBR	9.9	11.8	14.1	18.3	22.4	21.8	23.9	18.7	18.9	15.5	12.0	13.4		
		MCWBR	7.6	8.1	9.5	10.3	11.6	10.7	11.1	9.5	10.9	9.9	8.7	10.5		
		MCDBR	9.5	11.9	13.6	17.0	21.7	19.0	21.2	16.6	17.8	14.6	12.0	13.5		
	5% WB	MCWBR	7.7	8.4	9.4	9.9	11.5	10.0	10.3	9.2	10.7	10.0	9.0	10.7		
		MCWBR	7.7	8.4	9.4	9.9	11.5	10.0	10.3	9.2	10.7	10.0	9.0	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.201	0.257	0.291	0.319	0.330	0.330	0.341	0.329	0.300	0.279	0.211	0.179		
	taud		2.039	2.213	2.206	2.220	2.337	2.446	2.445	2.475	2.540	2.426	2.090	1.949		
	Ebn at Noon		152	224	258	274	281	284	277	271	260	218	143	92		
	Edh at Noon		10	19	29	35	34	31	31	27	21	16	10	6		
(44) All-Sky Solar Radiation	RadAvg		67	266	723	1335	1647	1601	1476	1147	716	320	93	35		
	RadStd		10	38	77	79	98	116	133	102	68	36	16	3		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-216	-259	N/A	N/A	

Nomenclature: See separate page